Learning and Assessment

This session explored how AI is reshaping learning and assessment in STEM. Faculty discussed how to maintain rigor and cognitive development while leveraging AI for personalization, feedback, and engagement.

Key Findings from Faculty Discussion

- **Employability & Preparedness:** Recent graduates with AI literacy report stronger job alignment and faster employment, but concerns remain over how long AI skill advantages will last before the lack of core disciplinary mastery becomes limiting.
- **Pedagogical Experimentation:** Assignments comparing student-generated versus AI-generated work reveal that students cannot meaningfully critique AI outputs without mastering underlying methods.
- **Cognitive Erosion:** Faculty noted visible differences in analytical depth when students rely heavily on AI for problem solving; raising the question of whether over-reliance bypasses mastery in STEM.
- **Assessment Redesign:** Ideas included incorporating reflection prompts ("What did you learn to do better?"), student-generated challenge questions, and low-stakes quizzes to measure process and growth.
- **Academic Integrity Reframed:** Faculty debated whether traditional notions of "cheating" remain relevant; some favored reframing integrity through redesigned learning objectives that emphasize reasoning and application.

References

Design and assessment of AI-based learning tools in higher education (2025) (link)
Structured and intentional AI integration can enhance personalization and feedback, offering scalable learning support for students while maintaining instructor oversight.

Evaluating AI-Powered Learning Assistants in Engineering Higher Education (2025) (link)
AI assistants improved engagement and comprehension among engineering students, with many reporting higher efficiency and deeper understanding of technical concepts.

Artificial intelligence skills and their impact on the future workforce (2025, PMC) (link)
Graduates with advanced AI literacy show stronger alignment with their fields and greater employability, highlighting the importance of AI competencies in higher education.

Talk is cheap: why structural assessment changes are needed for a time of GenAI (2025) (link)
Generative AI challenges assessment validity, enabling surface-level completion of tasks without genuine understanding — prompting a call for reimagined assessment practices.

The effects of over-reliance on AI dialogue systems on students (2024) (link)
Excessive dependence on AI chat systems can weaken students' cognitive and analytical skills, reducing self-reliance and critical engagement with material.

Scaffold or Crutch? Examining College Students' Use and Views of Generative AI Tools for STEM Education (2024) (link)
Many students rely on AI-generated solutions rather than reasoning through problems themselves, risking erosion of fundamental STEM problem-solving abilities.

LEARNING and ASSESSMENT

- graduates w/ AI skills are more likely to be hired → how long will this last

- graduates demand widespread adoption of A.I.

  Assignments where students do something (create code) and compare to A.I. solution
  Then - assignment where students only use A.I. and try to assess what A.I. did
  Notice the difference (i.o.w.: you can't critique A.I. without knowing how to do it)

- crutch effect?

- erosion of core cognitive skills? (Gerlich, 2025) Societies

- ~~bypassing~~ Bypassing mastery in STEM?

- how do you redesign assessment to capture <u>process</u> rather than answer?

  Reflection → What did you learn to <u>do</u>? What have you learned to do better? How are you growing?
  Challenge questions created by students
  Hands-on activities